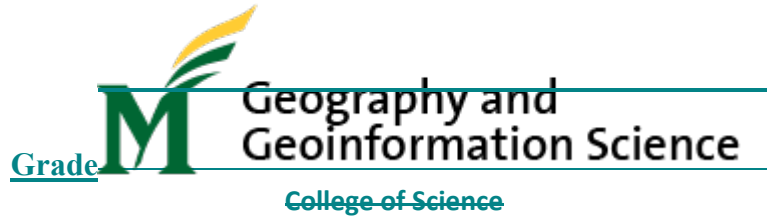


File Name: A7R The Fight for Education

Argument



**GGG-300: Quantitative Methods for Geographical Analysis**  
**Course Syllabus**

Associated Term: Spring 2025  
Levels: Non-Degree, Undergraduate, Consortium  
Attributes: Undergraduate—Upper Division  
Instructors: Donglian (Lillian) Sun (P)  
Fairfax Campus  
Lecture Schedule Type  
3.000 Credits  
CRN: 71719

**Course Instructor: Dr. Donglian (Lillian) Sun**

E-mail: [dsun@gmu.edu](mailto:dsun@gmu.edu)  
Office: Exploratory Hall #2407  
Phone: 703-993-4736  
Office hours: 3:00-3:30 PM on Monday or by appointment.  
Course Web Page: <http://courses.gmu.edu>

Class Location: **Exploratory Hall 2103**  
Class Times: 1:30 p.m. to 2:45 p.m. every Monday and Wednesday (in-person)  
Class Dates: 01/21-05/14, 2025

**Required Texts:**

McGrew, J.C., Lembo, A.J., and C.B. Monroe. 2014. An Introduction to Statistical Problem Solving in Geography (Third Edition). Waveland Press, Inc., Illinois. ISBN: 1478611197  
Make sure to get the Third Edition! Available at the GMU Bookstore or order online at [www.waveland.com](http://www.waveland.com)

**Other Requirements:** Flash drive/memory stick.

**Course Overview:** A survey of quantitative methods commonly used in geographic research. Emphasizes spatial analysis techniques.

Lab assignments will be based on the lecture material previously delivered and available as PowerPoints on Blackboard. Each lab assignment will be due one week after it is assigned (and at the start of the lecture). Late labs will only be marked for the usual documented medical reasons or by previous agreement with the instructor. Deployment of any family member is, of course, an acceptable reason for special arrangements to be made.

#### **Course Grading:**

Initial Test \_\_\_\_\_ 5%

(Each student will be awarded 5% for completing this test i.e. every student will get an A since I simply want to see what you know at the start of the course)

Attendance \_\_\_\_\_ 10%

Lab Exercises \_\_\_\_\_ 30%

Mid-term Exam \_\_\_\_\_ 25%

Final Project \_\_\_\_\_ 30%

All parts of the course are graded with a letter grade e.g. A+ B C etc. For the multiple choice tests and labs letter grades are assigned as follows:

**A+ 95% and over or top mark; A 91 to 94; A- 87 to 90**

**B+ 85 to 86; B 81 to 84; B- 75 to 80**

**C+ 67 to 74; C 64 to 66; C- 60 to 63**

**D+ 57 to 59; D 50 to 56; F less than 50**

#### **CLASS SCHEDULE (subject to change)**

**note: The lab dates below refer to the date they will be assigned!**

Schedule

#### **Lecture I**

**Introduction to the Course**

**Benchmark Test to Establish Student's Level of Knowledge**

McGrew, Ch 1: The Context of Statistical Techniques

#### **Lecture II**

**Characteristics of Geographic Data: Concepts**

McGrew, Ch 2

#### **Lecture III**

**Descriptive Statistics**

McGrew, Ch 3

#### **Lecture IV**

**Descriptive Spatial Statistics**

McGrew, Ch 4

Lab 1: Context for Statistical Analysis: Questionnaires and Surveys

#### **Lecture V**

**Probability**

McGrew, Ch 5

Lab 2: Data Presentation & Description with SPSS

#### **Lecture VI**

## **Sampling**

McGrew, Ch 6

Lab 3: Data Description Using SPSS (continued); Probability Theory

## **Lecture VII**

### **Estimation in sampling**

McGrew, Ch 7

**Range** Lab 4: SPSS, Normal Distribution; Standard Error of the Mean

## **Lecture VIII**

### **Elements of Inferential Statistics**

McGrew, Ch 8

## **Mid-term exam**

## **Lecture IX**

### **Two Sample and Matched Pairs Difference Tests**

McGrew, Ch 9

Lab 5: Chi-Square One-Sample Goodness-of-Fit Test; Two-Sample Difference Tests

## **Lecture X**

### **Three or More sample Difference Tests: Analysis of Variance**

McGrew, Ch 10

## **Lecture XI**

### **Goodness-of-Fit Tests and Categorical Difference Tests**

McGrew, Ch 11

Lab 6: Wilcoxon-Mann-Whitney Test for Two Independent Samples (using SPSS);

Chi-Square 2 to K-Sample Test (Contingency Table Analysis).

## **Lecture XII**

### **Inferential Spatial Statistics**

McGrew, Ch 13 and 14

Lab 7: One and Two-Way Analysis of Variance Using SPSS

## **Lecture XIII**

### **Correlation**

McGrew, Ch 16

## **Lecture XIV**

### **Regression**

McGrew, Ch 17

Lab 8: Correlation and Regression Analysis

## **Lecture XV**

### **Multiple Regression**

Cluster Analysis

McGrew, Ch 18

## **Lecture XVI**

### **Epilogue: Statistical Problem Solving in Geography**

Final Project

## **Student Resources:**

- **Academic Integrity:** Students must be responsible for their work, and students and faculty must take on the responsibility of dealing explicitly with violations. The tenet must be the foundation of our university culture. [See <http://academicintegrity.gmu.edu/distance/>].
- **Honor Code:** Students must adhere to the guidelines of the George Mason University Honor Code [See <http://oai.gmu.edu/the-mason-honor-code/>].
- **MasonLive/Email (GMU Email):** Students are responsible for the content of university communications sent to their George Mason University email account and are required to activate their account and check it regularly. All communication from the university, college, school, and program will be sent to students solely through their Mason email account. [See <https://masonlivelogin.gmu.edu/>].
- **Patriot Pass:** Once you sign up for your Patriot Pass, your passwords will be synchronized, and you will use your Patriot Pass username and password to log in to the following systems: Blackboard, University Libraries, MasonLive, myMason, Patriot Web, Virtual Computing Lab, and WEMS. [See <https://password.gmu.edu/index.jsp>].
- **University Policies:** Students must follow the university policies. [See <http://universitypolicy.gmu.edu>]. Responsible Use of Computing—Students must follow the university policy for Responsible Use of Computing. [See <http://universitypolicy.gmu.edu/policies/responsible-use-of-computing/>].
- **University Calendar:** Details regarding the current Academic Calendar. [See <http://registrar.gmu.edu/calendars/index.html>].
- **Students with Disabilities:** Students with disabilities who seek accommodations in a course must be registered with the George Mason University Office of Disability Services (ODS) and inform their instructor, in writing, at the beginning of the semester [See <http://ods.gmu.edu>].
- Students are expected to follow courteous Internet etiquette at all times; see <http://www.albion.com/netiquette/corerules.html> for more information regarding these expectations.

## **2. Student Services:**

- **University Libraries:** University Libraries provides resources for distance students. [See <http://library.gmu.edu/distance> and <http://infoguides.gmu.edu/distance-students>].
- **Writing Center:** The George Mason University Writing Center staff provides a variety of resources and services (e.g., tutoring, workshops, writing guides, handbooks) intended to support students as they work to construct and share knowledge through writing. [See <http://writingcenter.gmu.edu>]. You can now sign up for an Online Writing Lab (OWL) session

just like you sign up for a face-to-face session in the Writing Center, which means YOU set the date and time of the appointment! Learn more about the [Online Writing Lab \(OWL\)](#).

---

### **The Fight for Education**

*I ran as fast as I could. I could see my destination just ahead of me. Just a little further, I kept saying to myself, just a little further and I will have made it to the land of the free and the home of the brave, the land of opportunity.*

*Diego finally got to his destination. He had gotten to America without getting caught. Now he could get the education to become a mechanic.*

Like Diego, many illegal immigrants come to America every day for a better life and a better education. Whether these illegal immigrants should get the education they came for is very controversial in today's economy. Some Americans think that illegal immigrants shouldn't be eligible for instate tuition, but other Americans think that these illegal immigrants should be eligible for instate tuition. I am one who thinks that illegal immigrants should be eligible for instate tuition.

One of the most important arguments is that illegal immigrants pay taxes. By buying goods in America they are paying sales taxes, and in April most files and pay their [income] taxes. In fact, illegals pay about nine billion dollars in [Social Security] taxes every year. Where do these taxes go? They go to lowering the cost of instate tuition, making it so affordable today. So why shouldn't the illegal immigrants be eligible for instate tuition when they are helping to lower the price of the tuition?

Another important argument is one that our forefathers set forth to make a better nation. "We hold these truths to be self-evident, that all men are created equal...and hold unalienable rights, among these are life, liberty, and the pursuit of happiness." If we are truly equal, then illegal immigrants deserve the same rights as we do for instate tuition.

However, some Americans say that illegal immigrants don't deserve instate tuition because they are sending money back to their home country and families. But we do not know how much money they are actually sending back, and besides, this helps spread the money around. It may even help a foreign country to get out of poverty. This could actually stop more illegal immigrants from coming to America because they would

be stable enough from the money that is being sent to them so that they don't need to look to the land of opportunity for a job.

In conclusion, illegal immigrants should be eligible for instate tuition so, like Diego, they can follow their dreams. If you ever have to take a side on this issue, I hope you will consider mine.

**File Name: A 9-10R Is It Love?**

**Argument**

**Grade 9**

**Range of Writing**

### **Is It Love?**

One of the most misunderstood emotions is love. There are so many different perspectives that it can be interpreted as something completely opposite of its intention. Harper Lee, the author of *To Kill a Mockingbird* states that "it's a love story, plain and simple." This statement confuses many people because her book is anything but a romance novel. However, that is not the kind of love she is referring to. This is the love among family, friends, neighbors, and enemies. This love is called *agape*, and as Martin Luther King defines it, it is "a disinterested love in which the individual seeks not his own good, but the good of his neighbor...it begins by loving others *for their sakes*...it's a love seeking to preserve and create community" (King 19-20). In *To Kill a Mockingbird*, love is brought up in many ways. Love is shown throughout the book as different actions by the characters coming together so they can understand each other.

Atticus shows love towards his neighbor Ms. Dubose, even though she is completely disrespectful towards him. Everyday she insults Atticus and his children because he is defending Tom Robinson, but he does nothing to stop her. Instead, he sends Jem over to her house everyday for about a month to read to her. One day, it is Atticus who goes to her house, and he stays for most of the day. When he returns

home, he tells Jem that she has died, and the reason he had to read to her was because “Ms. Dubose was a morphine addict”(147). Atticus knew that she wanted to die free, and he helped her achieve her goal by sending Jem to read to her, so she could get her mind off the drug. “She said she was going to leave this world beholden to nothing and nobody” Atticus tells Jem (148). This is when Atticus and Jem realize her true determination to no longer be a morphine addict, and it’s in this way that they both truly understand her. Even though she was dying, Atticus still sends Jem to read to her everyday to help her become sober. Even when she is dying, Ms. Dubose is still trying to conquer her addiction, and it’s for that reason Atticus says, “She was the bravest person I ever knew” (149). Love is shown here because even though Ms. Dubose is completely disrespectful towards Atticus, he steps up to help her die free. This is an example of agape because instead of fighting hate with hate, Atticus loves Ms. Dubose, and by doing so, he is able to understand her, and therefore love her.

Another place love shows up is during the trial. Atticus truly believes that Tom Robinson is not guilty, and he goes out of his way to prove his point in courts. A majority of Maycomb is mad at Atticus for even taking the case, let alone actually trying to set him free. Atticus turns to the jury and tells them, “I am confident that you gentlemen will review without passion the evidence you have heard, come to a decision, and restore this defendant to his family”(275). Atticus fights as hard as he can to prove his case, but it just isn’t what people want to hear. The jury votes Tom Robinson guilty, and less than a week later he is shot. This upsets Atticus, and especially Jem. But Atticus comforts Jem by telling him it affected somebody on the jury, because one of the Cunninghams wanted to vote ‘not guilty’. Confused, Jem says, “One minute they’re tryin’ to kill him and the next they’re tryin’ to turn him loose... I’ll never understand those folks as long as I live” (298). Jem is too young to really understand, but Atticus knows his case is changing the views and thoughts of racism in their community. After the trial, Atticus is surprised to find that the entire black community gave his family a surplus of food as a thank you. This touches his heart because he starts crying, and he tell Calpurnia to “Tell them I’m very grateful” (86). We again see love in this situation because Atticus got the jury to understand Tom,

even though he ends up dead. Also, when Tom's friends and family reach out to thank Atticus for his efforts and his true support in the trial, we can see that there is understanding. One step at a time, Atticus is changing the thoughts of the people in the community to make it stronger by using agape.

One of the places we see the most love is with Boo Radley. He is just a misunderstood man who Jem and Scout torment and talk about throughout the summer. Boo, however, does little things to change their opinion of him. One night when there is a fire, Boo Radley comes out of his house and puts a blanket on Scout without knowing- a very loving and caring gesture. When Jem tells Atticus, he says, "Someday, maybe, Scout can thank him for covering her up" (96). After this, the kids start thinking of Boo differently. One night, after the play, Jem and Scout are attacked in the woods by Mr. Ewell when they are walking home. Suddenly, somebody kills their attacker and brings injured Jem home. It's not until they are home that Scout realizes it was Boo who saved them. "Hey, Boo" she says (362). This is the part of the story that the reader sees how much Boo Radley loves Jem and Scout. When Scout looks out from Boo's porch, she glimpses his perspective, and summarized the events that took place over the summer, and thinks, "Autumn again, and Boo's children needed him" (374). Boo cares for these children so much that he considers them his own kids. This is another example of agape, because he's giving without expecting anything in return. It's when Jem and Scout finally realize this that they know they love him back, which brings them closer together.

We see countless examples of love throughout the book through the characters actions. Martine Luther states that agape is "understanding" (King 19). The more the characters understand each other, the more they come to care and love each other. This makes them all, as a community, much stronger. "It's a love story, plain and simple" Harper Lee states. Not a romance, but a book in which agape is shown throughout. This is a book of understanding, and when there's understanding, there is agape. This love is important because this is what human nature needs in order to become united. King and Lee see this importance, and consider agape a tool used to bring people together to make a stronger community.

Works Cited

King, Martin Luther. "An Experiment in Love." *A Testament of Hope: The Essential Writings of Martin Luther King*. Ed. James M. Washington. San Francisco, CA: HarperCollins, 199. 16-20.

Lee, Harper. *To Kill a Mockingbird*. New York: Grand Central, 1982. Print.

---

- 
- **Counseling and Psychological Services:** The George Mason University Counseling and Psychological Services (CAPS) staff consists of professional counseling and clinical psychologists, social workers, and counselors who offer a wide range of services (e.g., individual and group counseling, workshops and outreach programs) to enhance students' personal experience and academic performance [See <http://caps.gmu.edu>].
  - **Family Educational Rights and Privacy Act (FERPA):** The Family Educational Rights and Privacy Act of 1974 (FERPA), also known as the "Buckley Amendment," is a federal law that gives protection to student educational records and provides students with certain rights. [See <http://registrar.gmu.edu/privacy>].

---

**Disclaimer:** Any typographical errors in this Course Outline are subject to change and will be announced in class. The date of the final examination is set by the Registrar and takes precedence over the final examination date reported by the instructor.